

High-Level Design

Technical proposal · solution architecture

PROBLEM STATEMENT

**Integrated Video Management & Analytics
Platform**

FOR

**Gujarat Police Hackathon Innovation Challenge
2026 · State Crime Records Bureau**

SUBMITTED BY

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SOURCE

docs/high-level-design.md

1. The problem, restated

Gujarat operates roughly 80,000 cameras across 26 departments, bought at different times from different vendors, running different VMS software. They do not interoperate. The state wants live feeds matched against watchlists — stolen vehicles, wanted persons, missing persons — with real-time alerts, across the infrastructure it already owns.

The obvious design is to ingest everything, run analytics on everything, and alert on every match. That design fails three times over, and the failures are arithmetic rather than opinion. Section 7 does the sums.

My answer is a **capability registry that acts as a control plane**: the system measures what each camera can actually deliver, and assigns analytics accordingly. Cameras are not equal, and a platform that treats them as equal wastes its capacity on cameras that cannot produce evidence while missing the ones that can.

2. What I found in the sandbox

Every claim in this section was measured, not assumed. Several contradict the published documentation, and each cost a day's worth of wrong assumptions to discover.

The catalogue is nearly empty. The Resources page states that `GET /api/ingest` returns "every camera with its id, location, codec, live status, stream properties, and all three URLs" and that "the catalogue is the contract". In practice `/api/ingest` returns 404. The real endpoint is `/cameras.json`, and every record contains exactly two fields:

```
{ "id": "cam01", "name": "01 Chiman bhai Bridge" }
```

No location. No codec. No resolution. No stream URLs. **A registry that depends on the catalogue cannot be built.** Mine measures instead, which is why the discovery cost me nothing.

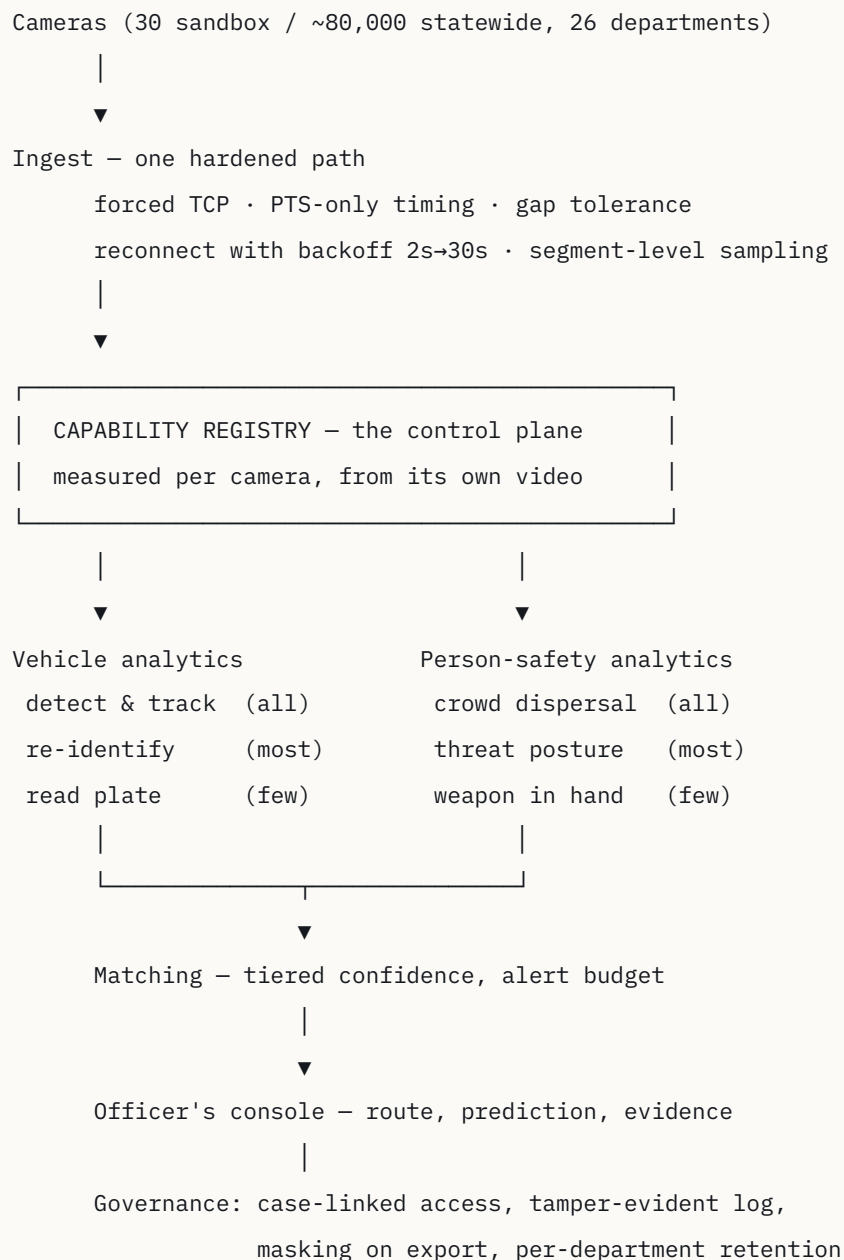
Only one transport is reachable. RTSP (8554) and WebRTC (8889) do not respond from the public internet; the origin sits behind a proxy that forwards only 80/443. HLS over HTTPS is the sole working path — which is also why the portal's own player shows a black frame.

The feeds are recordings, not live. Each camera serves a 12.00-hour AES-128 encrypted VOD of 7,200 segments carrying `EXT-X-ENDLIST`. The portal simulates liveness by seeking to `(now mod duration)`. Frames carry a burned-in stamp of 14 June 2026.

I have nonetheless built the client to be correct against live streams: the pre-submission checklist is met in full, and the 12–13 October test may well be live. Where a capability cannot be demonstrated against recordings — clock-drift measurement, for one — I say so rather than present a number that does not mean what it appears to mean.

Most of the grid cannot read a number plate. See section 4.

3. Architecture



The registry is the only component that decides what runs where. Every analytic is assigned; none is run blindly.

4. The capability registry

What is measured

Per camera, from its own video rather than its specification:

FIELD	WHY IT EXISTS
Delivered resolution	Declared resolution is frequently wrong
Real frame rate, from PTS	CAP_PROP_FPS disagrees with delivery
Vehicle scale → plate-pixel yield	The column that decides ANPR
PTZ pan share	A moving camera invalidates motion analytics
Night usability	Legible by day is not legible by night
Clock confidence	A wrong clock silently corrupts a timeline
Governance class	Health footage carries extra duty of care

The finding that shapes everything

Reliable plate reading needs roughly 120 pixels of plate width. All 30 government cameras were surveyed, sampled at 09:00 in each recording — daylight, median frame brightness 110 of 255. Nothing below was inferred from a specification.

CAMERA	RESOLUTION	PLATE P90	ANPR YIELD	VERDICT
cam18 Rajkot City	1920×1080	272 px	21.7 %	plate-capable
cam17 Rajkot Bus Port	1920×1080	201 px	31.2 %	plate-capable
cam09 New Bypass, Junagadh	1920×1080	193 px	54.6 %	plate-capable
cam05 Visat Teen Rasta	1920×1080	89 px	2.8 %	occasional
cam12 Adalaj Tollnaka	1280×720	68 px	5.4 %	occasional
cam22 BK Mervada Tran Rasta	854×480	63 px	0 %	presence-only
cam02 Janpath	1920×1080	63 px	1.6 %	presence-only
cam14 Delight RLVD	1920×1080	45 px	0.6 %	presence-only
cam16 Visat P2	1920×1080	22 px	0 %	presence-only
14 further cameras	mixed	≤ 48 px	0 %	presence-only
7 cameras	mixed	—	—	no vehicles observed

Three cameras in thirty can read a number plate. Eleven cameras are 1920×1080, and only three of those eleven are among them.

The clearest single comparison in the table: **cam16 is 1920×1080 and yields 22 px of plate. cam22 is 854×480 and yields 63 px** — a quarter of the pixels, three times the plate. Resolution predicts nothing. Mounting distance and angle predict everything, and neither appears on any specification sheet or in any procurement document.

Seven cameras observed no vehicles at all in the sampled window. They are not broken; they are pointed at places vehicles do not go. A watchlist platform that assumes every camera is a traffic camera wastes a quarter of the estate before it starts.

This is why capability on a real grid is a **yield**, not a yes/no — a vehicle directly beneath the mast is readable while the same vehicle two lanes over is not. The yield is the number that tells Gujarat Police where ANPR is worth deploying.

An earlier survey of this grid covered only 16 cameras and was sampled, unknowingly, at a position that was night-time in most recordings. It is superseded here rather than quietly amended: the ordering it reported — that mounting distance dominates resolution — survived re-measurement in daylight across the full thirty, but its figures did not, and the figures are what a reviewer would have checked.

Consequence for the live test

The organisers' test case asks entrants to **track a designated vehicle**, not to read its plate. On this grid those are different problems. An entry built on ANPR alone will demonstrate beautifully on a close-up and produce nothing on the government feed. Identity must therefore be carried by appearance — type, colour, size, direction, timing — with the plate serving as corroboration on the minority of cameras that can supply one.

5. Analytics, assigned by capability

Capability is not one number. A camera on a gantry over a carriageway can be excellent at number plates and useless for faces; a camera at door height in a ticket hall is the reverse. So the registry measures and stores the axes separately, and each analytic is assigned against its own.

CAMERA CAPABILITY	VEHICLE ANALYTIC	PERSON-SAFETY ANALYTIC
plate-capable	full ANPR + tracking	weapon detection
plate-occasional	ANPR as corroboration	threat posture
presence-only	detect, track, re-identify	crowd dispersal
no vehicles observed	presence detection	crowd dispersal
unusable	none	none

The person-safety lane deserves a note. A knife is roughly 20 cm; on a camera whose plates measure 27 px, a knife in a hand is about 8 px. It is not in the image, and weapon detection there is fiction. But **you do not need to see the weapon to detect the attack** — when a weapon appears in a public place, a crowd

disperses from a point, and that signal is large, model-free, and visible on the worst camera on the network. It turns a camera that can do nothing else into a violence sensor.

5.1 Facial recognition

The brief asks for an analytics approach covering ANPR, facial recognition and tracking. Facial recognition is designed into this architecture and assigned by the registry exactly as ANPR is — against a measured axis of its own.

The measurement. Face recognition is limited by inter-ocular distance, the pixels between the eye centres. ISO/IEC 19794-5 puts 60 px as the floor for an enrolment image; this system treats 40 px as the floor for even attempting a match, which is generous. Because running a face detector on cameras that cannot resolve faces would measure my own false positives, the scale is derived from person detections I already trust, through stated anthropometry — head height $\approx$ standing height / 7.5, head width $\approx$ 0.6 $\times$ head height, inter-ocular $\approx$ 0.33 $\times$ head width, composing to standing height / 37.9. The constant is published in `sentinel/facecap.py` so a reviewer can disagree with it and recompute.

The derivation is generous twice over, and both favour the camera: it assumes the subject is upright, fully visible and facing the lens. A seated or walking-away person yields less.

The result, on the cameras I hold footage for:

CAMERA	PEOPLE MEASURED	BODY HEIGHT P90	INTER-OCULAR	VERDICT
cam09 New Bypass, Junagadh	139	370 px	9.8 px	face-unusable
cam16 Visat P2, Ahmedabad	310	166 px	4.4 px	face-unusable
cam05 Visat Teen Rasta	343	113 px	3.0 px	face-unusable
cam01 Chimanbhai Bridge	945	101 px	2.7 px	face-unusable

The best camera on this grid is short by a factor of four. The worst is short by fifteen. At 2.7 px there is no face present — there is a smudge that resembles one, and a recogniser pointed at it does not fail quietly. It returns a name.

So no camera on this grid is assigned facial recognition, and the system will not run it. That is the registry doing its job, not a gap in the build: the same control plane that declines to schedule ANPR on 27 of 30 cameras declines to schedule face matching on all of them. Onboard a camera that clears 40 px — a ticket hall, an entrance lobby, a checkpoint at head height — and it is assigned automatically, with a ceiling that caps any face match at *probable* and never *confirmed*, because a registration is a registered fact whereas a face match is a similarity.

Two positions I hold deliberately, and will defend:

- **Person detection and tracking, yes; biometric identification, no — on this grid.** The system tracks people, measures crowd motion and reads limb geometry. None of that identifies anybody. The distinction matters legally as well as technically: tracking an unidentified person through a scene and matching a face against a gallery are different intrusions and attract different scrutiny.

- **No expression or emotion analysis anywhere, at any resolution.** Inferring intent from a face is contested science. Applied by a police system to the public it is a liability rather than a feature, and it is the first thing a defence would attack.

Scope of this measurement, stated plainly: four cameras, not thirty. The gateway returned HTTP 521 during the full-grid run, so the remaining twenty-six are measured on the next successful pass — `tools/survey_faces.py --grid` produces the table above. The conclusion is not in doubt at this margin, but four is what has been measured and four is what is claimed.

6. Matching, confidence and alert budget

Alert fatigue is what actually kills these deployments. Eighty thousand cameras matched against a national watchlist generate more alerts in an hour than a control room reads in a week, and once officers begin dismissing alerts reflexively the system is worse than useless — worse, because everyone still believes it is working.

A match therefore returns a **tier**, never a boolean:

TIER	MEANING	BEHAVIOUR
confirmed	capable camera, clean read	raises an alert
probable	marginal camera, or inexact plate	raises a lower-priority alert, with the doubt shown
corroborating	too weak to stand alone	strengthens an existing track; never notifies

Three things demote a match, and each is recorded so the officer sees why: an inexact plate, a camera not measured capable of reading plates, and a camera whose own clock is not trusted.

Plate matching is confusion-aware. OCR reliably confuses `0/0` , `5/S` , `8/B` . A degraded read of `GJ05MN8899` returned `GJ0SMNS899` — every wrong character a shape collision. Ordinary matching calls that a different vehicle and the stolen car is missed, on precisely the cameras where a miss matters most. Those substitutions carry reduced cost, so the vehicle surfaces as an inexact match, demoted a tier, rather than vanishing.

The alert budget is explicit — a stated ceiling per control room per shift. When it is spent, lower tiers stop notifying and become corroboration. Confirmed alerts always pass.

7. Scaling to 80,000 cameras

The full treatment -- hardware and software requirements, hot/warm/cold retention, load balancing and horizontal scaling, high availability and disaster recovery, estimated implementation and operational costs, the statewide rollout plan and the department-wise information requirements -- is in the

companion document `scalability-plan.md`. This section carries the arithmetic the architecture rests on.

The arithmetic below uses figures measured on this project, not vendor claims.

Analytics. An NVIDIA A100 sustains 40-60 simultaneous 1080p/30fps detection streams in published reference architectures. Running continuous analytics on all 80,000 cameras therefore needs on the order of **1,600 datacentre GPUs** -- expensive rather than impossible, but an order of magnitude beyond what a state video programme budgets for compute.

(An earlier draft of this document put the figure at 264,000 GPUs. That was extrapolated from a laptop GPU at 0.33s per frame and overstated the requirement by roughly 150x. It is corrected here rather than quietly dropped, because a number nobody can reproduce is worth less than no number.)

What real deployments do instead is exactly what this design proposes: they do not run heavy analytics everywhere.

These figures are indicative, drawn from vendor and press descriptions of each programme rather than from published operating data, and I have not been able to source primary references for them. They are included to show that selective assignment is ordinary practice, not to carry the argument — the argument rests on the measured share of this grid, below, which is reproducible from this repository.

DEPLOYMENT	CAMERAS	SHARE RUNNING FULL ANALYTICS
Port of Rotterdam	5,000	15 %
Rio de Janeiro (COR)	10,000	40 %
Aptibit, India	10,000	30-40 %
Changi Airport	2,000	70 % (high-risk site)
Beijing Traffic Bureau	8,000	20 % real-time, rest batched overnight

Selective assignment is not a novel idea; it is universal practice. **What differs here is the basis for the selection.** Those deployments tier cameras by human judgement -- importance of the location, risk of the zone, time of day. This design tiers them by *measurement*: what the camera can physically resolve, established from its own video. A camera at a critical junction that delivers 45px of plate will be given plate reading by a zone-based policy and will never produce a usable read. Mine will not be asked.

On the measured distribution of this grid -- 3 cameras in 30 able to support plate reading -- the heavy tier lands at 10 %, which is below every figure in the table above and is a measured number rather than an assumption. Including the two occasional-yield cameras raises it to 17 %, still below all but one.

Bandwidth. A 1080p H.264 stream is 2-5 Mbps. Centralising all 80,000 is 160-400 Gbps sustained. Published deployments avoid this by sending metadata rather than video to the centre, a 95-98 % reduction, with video held in a rolling buffer at the edge and pulled on demand for an investigation. The registry makes that tractable: it says which cameras justify an edge inference node at all.

Storage. 4 Mbps continuous is about 43 GB per camera per day -- **3.4 PB per day** statewide, 103 PB at 30-day retention. Full retention of raw video is not viable. The design retains metadata and sightings indefinitely, evidence clips per case, and raw video only for a short rolling window, per department policy.

The organisers' own constraint proves the point. Their Resources page states that each connected client receives its own copy of the stream and instructs entrants to open only the cameras they are actively processing. That is the 80,000-camera problem stated in miniature, in their own documentation. Something must choose which cameras to open. Mine chooses on measured evidence.

8. Integration strategy: heterogeneous cameras, NVRs and VMS

Twenty-six departments bought at different times from different vendors. The integration layer therefore has to assume nothing about what it is talking to, and has to keep working when what it is talking to changes -- which it did, mid-project, when the sandbox catalogue moved endpoint and lost every field except the camera id.

The adapter ladder

Each camera is onboarded by the highest rung it supports. Nothing is required to support more than rung 4.

RUNG	INTERFACE	WHAT IT GIVES	USED WHEN
1	Vendor VMS API / SDK	Streams, PTZ, events, recorded playback	Department runs a supported VMS and will share an API account
2	ONVIF Profile S / T / G	Stream discovery, PTZ, recording playback	Camera or NVR is ONVIF-conformant
3	RTSP (TCP) or HLS direct	Live stream only	Endpoint reachable; no management API
4	NVR file pull	Recorded segments only	No live path; forensic use only

The connector framework is a plugin interface: a new vendor is a new adapter implementing discover / open / close, not a change to the platform. Departments keep operational control of their own VMS throughout -- this is federation, not migration, which is why Model 4's rip-and-replace is rejected on cost and lock-in grounds.

What the platform refuses to depend on

The sandbox taught this the hard way. The documented catalogue endpoint began returning 404 mid-project; the replacement carries `{id, name}` and nothing else -- no location, codec, resolution or stream URL. A platform that reads camera capability from a catalogue could not have been built against that grid at all.

So the contract is deliberately thin: **an endpoint and an owning department**. Everything else -- resolution, delivered frame rate, codec, what the camera can actually resolve -- is measured from the video. One sandbox camera declares `r_frame_rate=250/1` in its container while reporting 29.96 in another field of the same file; the measured delivery is 29.98. Believing either header would put every sighting in the wrong second.

Correlating with watchlist databases

The watchlist is a store of entities -- stolen and blacklisted vehicles, wanted and missing persons, suspect lists -- normalised on onboarding so that comparison is confusion-aware rather than exact: a plate read as `GJ0IAB1Z34` still matches `GJ01AB1234`, and is demoted for having needed the recovery.

For deployment the authoritative sources are departmental rather than mine:

SOURCE	CARRIES	INTEGRATION
VAHAN	Vehicle registration, ownership, status	Pull for verification on a confirmed read; never the primary match path
SARATHI	Driving licence	Case linkage only
eGujCop	FIR, case records, officer identity	Case reference validation; the natural home for authentication and RBAC
CCTNS	Crime and criminal records	Watchlist population
AFIS / NAFIS	Biometric	Out of scope: this grid cannot resolve a face (§5.1)

Correlation runs continuously against the local watchlist replica so an alert never waits on an external system. VAHAN is consulted *after* a match to enrich it, not during, because a national registry must not sit in the latency path of a street-level alert.

9. Deployment and cybersecurity architecture

Deployment tiers

Three tiers, sized in the scalability plan and summarised here.

Edge decodes, detects, tracks and discards. It forwards metadata and evidence clips -- 2-5 % of raw volume -- because centralising 80,000 streams is 160-400 Gbps sustained and is not available at any budget.

District aggregates, runs the assigned heavy analytics, correlates against the watchlist, raises alerts, and holds 72 hours of hot video. Districts operate autonomously under network partition and reconcile on restore.

Central holds the registry, the statewide search index, case management, the audit chain, and warm/cold archive. It is the only tier that must serialise writes, because the audit chain cannot fork.

Security controls

CONTROL	STATE
TLS on every hop; RTSP forced over TCP	Built
Credentials environment-only, never committed	Built
Case-linked access, refused before a row is read	Built and tested at the HTTP layer
Tamper-evident hash chain over every query and export	Built and tested
Authentication against departmental identity	Designed, not built
Role-based access control	Designed, not built
Masking of uninvolved parties on export	Designed, not built
Network segmentation; camera VLANs isolated	Design

The roles the design assumes: **officer** (search within own case), **supervisor** (authorise, review alerts), **auditor** (read the chain, not the footage), **administrator** (onboard cameras, never search).

Stated plainly, because a reviewer will find it: **the case-linked gate is real and tested, but there is no authentication behind it**. It trusts the officer name it is handed. In deployment it must sit behind eGujCop or departmental AD with MFA on export and watchlist edits. A gate without an identity behind it is a control on paper.

That gap was found by auditing this codebase before publication, along with a worse one: a route returning full movement history with no credentials at all, whose audit write was conditional on the credentials that were missing. It is fixed, and there is now a test that enumerates every route and fails if one is left ungated.

10. Future roadmap

HORIZON	WORK	WHY IT IS NEXT
Pre-finale	Authentication + RBAC; masking on export; retention enforcement	The three designed-not-built controls above. All are named in the brief.
Phase 1-2	Statewide capability survey; lens-first optimisation	The 10 % plate-capable share is measured on 30 cameras. Eight of those 30 are one lens from capability.
Phase 3	Person re-identification in production; crowd-dispersal as an operator surface	Serves the wanted/missing-person watchlist categories that nothing currently serves.
Phase 4	VAHAN / CCTNS live integration; road-network routing for look-ahead	Replaces straight-line reachability with real routes.
Ongoing	Continuous re-survey and capability-drift alerting	A camera that was plate-capable and is now presence-only has been knocked or obscured. Conventional monitoring reports it healthy.

Explicitly **not** on the roadmap: facial recognition on cameras that cannot support it, and any affect or emotion inference at any resolution. Both positions are argued in §5.1 rather than left as omissions.

11. Governance is architecture

Not a policy appendix — enforced in the system:

- **Case-linked access.** Every movement query carries a case reference and an authorising officer. Requests without them are refused before a row is read, and the refusal is itself logged.
- **Tamper-evident audit.** Each log entry hashes the previous one. An edited or deleted entry breaks the chain and is detectable. Verified by test: altering one row identifies exactly which.
- **No biometric face matching**, as a stated position with its reasoning, and enforced by measurement rather than by policy (section 5.1).

Designed, with the schema in place, but **not built** — listed here as outstanding rather than described as if they exist:

- **Masking on export.** Uninvolved parties should be obscured in evidence bundles. The export path does not do this today. It is the single largest gap between this document and the code.

One narrower control *is* built, and it is not the same thing. Where an image is published rather than exported — the stills the officer's console shows beside each sighting — the publication step works from an allow-list drawn from the sightings that survived withholding, so a row removed for privacy cannot leave its picture behind. That keeps one specific person out of a public URL. It does not obscure anybody inside an image a case is entitled to, which is what masking means and what is still missing. - **Per-**

department retention. The registry carries a `governance_class` per camera, and health footage is intended to attract patient confidentiality above ordinary privacy duty. No retention logic acts on that column yet, and on the sandbox grid every government camera is currently classed `open`.

Licensing. Every component is Apache-2.0 or BSD — OpenCV, Tesseract, torchvision. Ultralytics YOLO, the default choice for work of this kind, is AGPL, which in a state deployment is a procurement problem rather than a footnote.

12. What is proven, and what is designed

Stated explicitly, because the next stage is production feeds in front of the State Crime Records Bureau and anything overstated now becomes a problem on 12 October.

Proven, with tests (153 passing): capability measurement, agreeing to 2.6 % against known ground truth; clock-drift detection, which caught a deliberate 7-minute error to within 1 second; tamper-evident audit; tiered matching and the capability ceiling; cross-camera re-identification; look-ahead camera nomination; the officer's console; and that every route returning movement data refuses without a case reference and an authorising officer.

That last one is in this list because it was false until it was audited. `/api/officer/route` returned a citizen's full movement history to any caller with no credentials, and the audit write beneath it was conditional on the credentials that were missing — so the only unlogged reads were the ones that should never have happened. Four of my own tests called it unauthenticated and asserted on the results, which is how a suite ratifies a hole instead of finding it. It is now gated, and swept at the HTTP layer by a test that enumerates the application's own routes so the next one cannot be missed the same way.

The eight pre-submission checklist items are covered by automated tests, including reconnect-on-feed-restart, which cannot be tested against a government feed at all. **Those tests skip unless the local mock grid is running**, so a default `pytest` run reports them as skipped rather than passed. They are run with `grid/run.sh up`.

Measured on the real grid: camera capability and ANPR yield for all 30 cameras, no failures; transport and catalogue behaviour; per-frame inference cost.

Designed and demonstrated in simulation, not yet proven on real footage: the three person-safety layers. I have no real footage of a crowd dispersing or a weapon being drawn, and will not claim otherwise.

13. A note on method

Five times during development, a component produced a confident wrong answer that only revealed itself when the output was inspected rather than trusted: a clock read as two years out; a phantom 19-second drift from a single bad sample; a road sign measured as a 304-pixel number plate; one vehicle

counted where there were dozens; and an entire camera survey conducted, unknowingly, on night footage.

Every one was caught by looking at pixels instead of believing a number. That experience is the argument this submission makes about the product itself. A platform that can state how it knows something, and under what conditions, is worth more to a police force than one that merely answers — because the failure mode of a confident wrong answer is an officer acting on it.